

MICROSERVICES

Traefik Reverse Proxy

The dynamic, cloud-native application gateway & reverse proxy designed to
configure itself instantly with zero reloads or downtimes.

Architecture Basics

A look under the hood of Traefik's service discovery and routing engine.



The Core Routing Flow



Dynamic Service Discovery

Orchestrator Integration

Traefik continually polls orchestrator APIs to auto-discover containers and pods as they scale up or down.

- **Docker Labels:** Routing rules are declared directly on container metadata with zero static configuration files.
- **Kubernetes Ingress:** Out-of-the-box support for standard Ingress resources to manage classic HTTP routes.
- **IngressRoute CRDs:** Custom resources exposing Traefik-native features like TCP routing, middlewares, and SSL configurations safely.

`docker-compose.yml`

```
services:
  web-app:
    image: nginx:alpine
    labels:
      - "traefik.enable=true"
      - "traefik.http.routers.app.rule=Host(`app.dev`)"
      - "traefik.http.routers.app.entrypoints=websecure"
      - "traefik.http.services.app.loadbalancer.server.port=80"
```

Flexible Entrypoints



HTTP Entrypoint

Configured statically on port 80.
Listens to basic insecure queries
to forward or force automatic
HTTPS upgrading rules.



HTTPS Entrypoint

Configured statically on port 443.
Manages incoming SSL/TLS
certificates, cipher negotiations,
and ALPN matching.



Direct TCP/UDP

Custom port mapping directly
bypasses application protocol
layers, proxying binary streams to
internal database engines.

Advanced Middleware Engine

-  **Authentication Gates**
Easily secure applications using BasicAuth, centralized LDAP, or modern ForwardAuth delegate mechanisms.
-  **Rate Limiting & Safeguards**
Set absolute thresholds via RateLimit and prevent backend saturation using InFlightReq limitations.
-  **Filtering & Modifiers**
Filter connections with IPAllowList, strip path prefixes via StripPrefix, and override response headers on-the-fly.

Automated Security

Configuring instant SSL/TLS generation and automated renewals with zero manual overhead.



| ACME & Let's Encrypt

SSL Certificates

Traefik natively integrates with ACME directory services to automatically generate, install, and renew secure SSL certificates.

- **Dynamic Updates:** Certificates update in local storage or Kubernetes secrets without gateway service restarts or downtime.
- **HTTP-01 Challenge:** Fully automated verification resolving directly over standard web ports.
- **DNS-01 Challenge:** Connects with global DNS providers to obtain secure wildcard certificates (e.g., *.domain.com).

Layer 4 TCP Direct Routing

No HTTP Dependency

Proxy databases like **PostgreSQL**, **Redis**, and raw binary protocols instantly at Layer 4 (L4) instead of parsing L7 HTTP headers.

Enables low-level packet switching with high performance margins and lower overall memory usage.

L4 TLS Pass-Through

Route TLS-encrypted traffic based on the **Server Name Indication (SNI)** extension header without cracking open the certificate.

Maintain complete zero-knowledge security models directly from the client endpoints to internal database instances.

Extensible Plugin Ecosystem



Yaegi Go Engine

Powered by Yaegi, an embedded Go interpreter. Run standard, uncompiled Go code dynamically on-the-fly inside the routing pipeline with zero binary bloating.



Custom Middleware

Interpose custom middleware logic seamlessly. Modify headers, rewrite payloads, inject proprietary authorization flows, or enforce corporate security checks.



Dynamic Providers

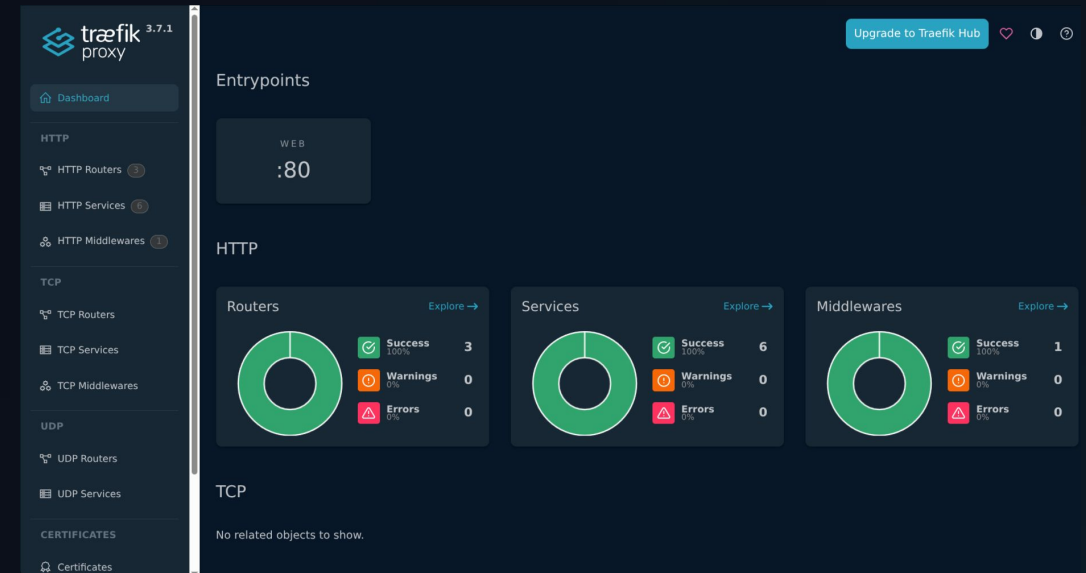
Extend service discovery capabilities. Query custom metadata servers, service meshes, or external databases to build configuration trees dynamically.

Traefik Dashboard & UI

Observability at a Glance

Traefik provides a robust, real-time graphical interface to inspect and monitor the cluster configuration instantly.

- **Visual Configuration Tree:** Easily trace network flows, verifying which Routers route to which Middlewares and Services.
- **Dynamic Status Checking:** Instantly spot unhealthy backends, missing ingress configurations, or misconfigured routing rules.



Questions & Answers

Thank you for your attention. Harness the full power of Traefik's modern,
cloud-native gateway architecture.

Learn more: doc.traefik.io